

Week 1.12 — 690 Grams — Buoyancy Goes from Arithmetic to Measurement, and the Electrical Track Picks Its Language

Sunday, September 6, 2026 · 3:00 – 5:00 PM · 11 students

Eleven of us at 1.12 — Trenton, Rhylan, Maddox, Liam, Isaac, Luke, Rebecca, Frankie, Oliver, Eli, and Tyler — from 3:00 to 5:00. Almost everything on the table this week was a promise from 1.11 coming due. The headline is buoyancy again, but a different kind of buoyancy than last time. At 1.11 we calculated the displacement of the PVC we had on hand and designed around the number. This week we cut it, built it, and put it in water. Two 1-inch PVC buoyancy tubes, 455 mm end to end, 285.6 grams and 285.8 grams — two parts that were supposed to come out identical came out identical, which is its own kind of good news. Both floated. Then we took the measurement that actually matters: the ROV weighs 3,500 grams sitting on the bench and 690 grams once it is in the water. That second number is what a buoyancy design is aimed at, and now we know it instead of guessing at it. The electrical track closed its 1.11 action item too, and closed it properly — KiCad is the team's schematic and PCB tool, chosen with the pros and cons written down, and ANSI is the symbol standard, chosen after comparing it against IEC and NEMA. With a tool and a notation settled, they drew the first outline of how power actually flows through the TriggerFish control box. The sensor and software track went after the float: what a buoyancy engine is (a syringe that pulls water in to sink and pushes it out to rise), what it requires (a watertight control box and a pressure sensor), and how a vehicle knows which way is up (a 9-DoF IMU on the Raspberry Pi's I2C bus, Bosch or Adafruit, about twenty dollars). Mechanical kept the TriggerFish frame moving in CAD and sketched manipulator concepts. Management put the TVA grant into the season budget. And marketing started building the slide show for STEAM night at Rainbow Elementary.

11

STUDENTS AT THE BENCH

2

PVC BUOYANCY TUBES BUILT AND
FLOAT-TESTED

690 g

THE ROV'S WEIGHT IN WATER —
AGAINST 3,500 G IN AIR

3

SYMBOL STANDARDS COMPARED
BEFORE CHOOSING ANSI

WHAT WE LEARNED

- Weight in air and weight in water are two different numbers, and the second one is what you design for. The ROV is 3,500 grams on the bench and 690 grams in the water — the water is already holding up most of it.
- Two parts that are supposed to be identical should measure identical. 285.6 grams and 285.8 grams out of the same build is evidence the process is repeatable, and repeatable is what you need when you build the second one and the third.
- A float test is cheap, fast, and final. You can argue with a calculation. You cannot argue with a tube that floats.
- Free is not the same as easy. KiCad costs nothing and gives you an enormous parts library and unlimited board space, and it is genuinely hard to learn and sometimes hard to find a part in. Writing both columns down is what turns a preference into a decision.
- There is more than one correct way to draw a resistor. ANSI, IEC, and NEMA are all real standards. The team picked ANSI because it already reads those symbols — and the choice matters mostly because everybody has to draw the same way.
- A buoyancy engine is a syringe. Draw water in and the float gets heavier and sinks; push it out and the float gets lighter and rises. The hard engineering is the watertight box around it and the pressure sensor that tells it how deep it is.
- An IMU is how a vehicle knows which way is up. Nine degrees of freedom gives you yaw, pitch, and roll, it talks to the Raspberry Pi over I2C, and it costs about twenty dollars — a lot of capability for the price of a pizza.

DECISIONS MADE

- KiCad is the team's schematic and PCB tool, and the reasoning is on the record: it is free, the parts library is enormous, and the board space is unlimited. It is also complex, and some parts are hard to find. That is a trade the team made on purpose instead of defaulting into.
- ANSI is the team's electrical symbol standard. ANSI, IEC, and NEMA were all compared, and ANSI won because those are the symbols the team already recognizes. Which standard matters less than everybody drawing the same way — a documentation package that mixes notations is one the judges have to decode.
- Buoyancy is built and measured now, not calculated. Two 1-inch PVC tubes at 455 mm, 285.6 g and 285.8 g, both positively buoyant in the water test — and the vehicle they have to lift weighs 690 g in water against 3,500 g in air.
- The float's requirements get written before the float gets designed. A watertight control box and a pressure sensor are not optional, and the buoyancy engine is a syringe that takes water on to sink and pushes it out to rise.
- An IMU goes on the short list — 9 degrees of freedom, on the Raspberry Pi's I2C bus, with both Bosch and Adafruit offering one around twenty dollars. Yaw, pitch, and roll is how the vehicle knows its own orientation.
- The TVA grant is in the season budget. Money that has been awarded goes on the budget sheet, and the next question the CFO put to the team is what we should prototype — so the prototypes get costed instead of improvised.

ACTION ITEMS

Bring candidate company names to the team for a vote (carried from 1.11)

Trenton (CEO) · by 1.13

Finish the integrated master schedule — every sub-team on one calendar (carried from 1.11)

Annabelle (COO) · by 1.13

Decide what the team should prototype so it can be costed into the budget

Maddox (CFO) · by 1.13

Fix the basic frame CAD

Adrian (Mechanical) · by 1.13

Keep the manipulator brainstorm going — a pros-and-cons list for each way of moving the claw

Eli (Payload) · by 1.13

Research the sensor modules and bring back a recommendation

Rhylan (Sensor) · by 1.13

Finish the STEAM night slide show for Rainbow Elementary

Lillian (Marketing) · Sept 30

Create the team Instagram account

Lillian (Marketing) · by 1.13

Get better photographs of the ROV for the marketing display

Lillian (Marketing) · by 1.13

STUDENT LEARNING OUTCOMES

- Cut, assembled, and weighed two 1-inch PVC buoyancy tubes to a common spec, then float-tested both.
- Measured the ROV's mass in air and in water, and understood why the in-water number is the design target.
- Evaluated KiCad against the team's needs and wrote down the pros and cons before committing to it.
- Compared the ANSI, IEC, and NEMA electrical symbol standards and chose one for the whole team to draw in.
- Drew a first outline of how power flows through the TriggerFish control box.
- Researched vertical profiling floats and buoyancy engines, and turned that research into requirements instead of a wish list.
- Researched 9-DoF inertial measurement units for the Raspberry Pi, including how they interface and what they cost.
- Advanced the TriggerFish frame in CAD and sketched payload manipulator concepts.
- Started the outreach slide show for STEAM night at Rainbow Elementary.

LOOKING AHEAD

Phase 1 still owes itself a closing — the final TriggerFish demonstration for parents, stakeholders, and sponsors, and the role retrospective that goes with it. Four items have now carried across two sessions and should close first: the company name vote, the integrated master schedule, capsule dive #3 (the gate is three consecutive dry dives), and the first walkthrough of a System Integration Diagram. The rest of the near-term work is concrete — the frame CAD gets finished, the manipulator concepts get a written pros-and-cons list, the sensor track brings back a recommendation, and the STEAM night slide show has a hard date: September 30.

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